PortFlow Container Terminal: Yard and Gate Flow

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Recommendation. Option B — Integrated flow package: shared visibility, appointments, AI-assisted yard planning — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,000,000 of the CAD $10,000,000 budget: NPV CAD $6,491,191 at 9.5%, payback 31.7 months. Five-year net financial impact (case formula): -CAD $16,500,000 (net savings).

Capex	NPV @ WACC	IRR	Payback	Benefit/ Cost	Run-rate benefit	Levelized cost
CAD $9,000,000	CAD $6,491,191	29.9%	31.7 months	1.46	CAD $6,400,000/ yr	128752.43 CAD per year per % point of improvement

1. Problem diagnosis

The yard problem is a dwell-time problem, not a space problem. Little's Law: 3,520 TEU/day x 6 days = 21,120 TEU (88% of 24,000). At +20% volume (4,224 TEU/day) the same 6-day dwell needs 25,344 TEU, 106% of capacity. Even with the 15-acre parcel (28,000 TEU) utilization would be 90.5%, so space alone cannot meet the 85% objective. Dwell must fall to 4.83 days or less (-19.5%).

- Core issue (cause): fragmented release and schedule information. Release status, vessel changes and pickup plans are not shared across teams, truckers, rail or customers, and planning runs in separate systems on fixed rules.
- It drives dwell: customers learn late that a container is released, so pickups slip and boxes sit in the yard.
- It drives truck queues: with no appointments or readiness information, trucks arrive in concentrated windows and queue 40 minutes.
- It drives rehandles: the system knows the stack but not the exact position, and stacks are not built around known pickup times, so crane time goes on reshuffles and trucks spend 32 minutes inside.
- Symptoms, not causes: yard crowding, $6M overtime, $8M penalties, customers unable to plan and staff anxiety. More land or more overtime treats these without removing the cause.

2. Options and decision

Opt	Alternative	Capex	Gain	In budget	Meets target	NPV	Payback	Score /10
A	Gate quick win: truck appointments and release portal	CAD $4,100,000	10%	yes	no	CAD $4,157,625	28.5 months	5.63
B ✔	Integrated flow package: shared visibility, appointments, AI-assisted yard planning	CAD $9,000,000	21.7%	yes	yes	CAD $6,491,191	31.7 months	7.0
C	Yard expansion on the 15-acre parcel with two added RTGs	CAD $14,500,000	14.3%	NO	no	-CAD $14,627,035	> 60 months	2.9

Criteria and weights: Target attainment 30%, Financial return (NPV) 25%, Payback speed 15%, Downside resilience (-15% perf.) 15%, Budget headroom 15%. Budget is a hard gate.

3. Five-year cash flow (CAD)

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
-CAD $9,000,000	CAD $620,000	CAD $5,100,000	CAD $5,100,000	CAD $5,100,000	CAD $5,100,000

4. Scenarios

Scenario	Gain	Capex	NPV	IRR	Payback	Meets target
Bear (-15% perf., +20% capex)	18.45%	CAD $10,800,000	CAD $620,445	11.3%	45.2 months	no
Base (approved plan)	21.7%	CAD $9,000,000	CAD $6,491,191	29.9%	31.7 months	yes
Bull (+15% perf., -10% capex)	24.95%	CAD $8,100,000	CAD $10,962,775	44.7%	25.7 months	yes

5. Sensitivity (NPV tornado)

Driver	Low NPV	High NPV	Swing
Annual benefit: achieved gain x unit value (+/-20%)	CAD $2,394,628	CAD $10,587,753	CAD $8,193,125
O&M cost (x2 / x0.5)	CAD $1,499,569	CAD $8,987,001	CAD $7,487,432
Capital cost (+20% / -20%)	CAD $3,692,866	CAD $9,289,515	CAD $5,596,649
Implementation duration (+50% / -25%)	CAD $4,623,006	CAD $7,514,022	CAD $2,891,016

6. Implementation roadmap

Horizon	Timing	Workstream	Budget	KPI	Gate
Phase 1	Months 1-3	Co-design and pilot	CAD $2,000,000	Pilot lanes: queue time -25%; release status shared within 1 hour; baseline rehandles measured	Slot adherence >= 80% on pilot lanes and no safety incidents
Phase 2	Months 4-9	Scale appointments, positioning and AI advice	CAD $5,200,000	Dwell <= 5.3 days; truck turn <= 55 min; rehandles per box -20%	AI advice accepted >= 70% with every override logged; tariff announced with 90 days notice
Phase 3	Months 10-12	Stabilise, connect rail and customers, hand over	CAD $1,800,000	Dwell <= 4.8 days; truck turn < 45 min; utilization on track for <= 85% at year 5	Finance confirms savings against each owner's target; contingency released

7. Risk register

Risk	L	I	Mitigation	Trigger
Truckers do not adopt appointments	Medium	High	Slot rules co-designed with trucking firms, grace windows, priority lanes for on-time trucks, a walk-in lane kept during transition	Slot adherence < 75% for 2 weeks
Dwell does not fall enough	Medium	High	Free-time tariff, release alerts, empties evacuation agreements; parcel design kept ready as phase 2	Dwell > 5.3 days at month 9
Cutover disrupts live operations	Medium	High	Lane-by-lane and block-by-block rollout, parallel running, go-lives outside vessel peaks, rollback plan	Gate throughput below baseline for 2 shifts
Staff resistance and role concerns	Medium	Medium	Staff involved before tools arrive, paid training, no layoffs from this program, redeployment into the planning cell	Training completion < 90% or AI advice acceptance < 50%
AI advice wrong or data quality poor	Low	High	Advisory only, 6 weeks in shadow mode, hard safety rules locked, weekly model review	Override rate > 30% or any poor miss linked to a recommendation

8. Judge defence

Q: Why not just build on the 15-acre parcel?
Space alone fails the objective: 25,344 TEU / 28,000 TEU = 90.5% at year 5. It costs about $14.5M, over the $10M budget, and must be phased around live operations. Cutting dwell to 4.7 days reaches about 83% on today's footprint. We keep the parcel as a phase-2 option if dwell stays above 5.6 days.

Q: What if dwell does not fall as much as you assume?
The threshold is explicit: dwell at or below 4.83 days meets 85% without land. If dwell is above 5.3 days at month 9 (our gate trigger), we tighten the free-time tariff and start the parcel design. With dwell at 5.6 days or less, the parcel would then meet the objective.

Q: How confident are you in the $6.4M of annual savings?
Each saving has an owner and a mechanism. Break-even needs only $3.1M a year. If penalties fall 25% instead of 50%, the five-year impact is still -$6.5M.

Q: Will the AI and automation cost jobs?
No layoffs from this program. The AI recommends and planners decide. Roles shift from manual re-planning and overtime to planning-cell and exception work, with training in paid time.

Q: How do you keep the terminal running during the rollout?
Lane-by-lane appointments with a walk-in lane kept open, block-by-block positioning, 6 weeks of shadow mode for the AI, go-lives outside vessel peaks, and a rollback plan triggered if gate throughput drops below baseline for 2 shifts.

Q: How will you get truckers to use appointments?
We co-design slot rules with them, guarantee the container is ready when a slot is booked, give on-time trucks a

priority lane, and keep grace windows. Slots exist only for released containers, so a booking is never wasted.

9. Assumptions

Parameter	Value	Source
Baseline	105.6 % of capacity	case datum
Target improvement	19.5%	case datum
Capital budget	CAD $10,000,000	case datum
Timeline	12 months	case datum
Option costs & gains	see options table	case datum
Monetisation	option-level annual savings stated in the case pack	assumption: estimate (overtime -40%, penalties -50%; dwell 6.0 to 4.7 days)
Discount rate (WACC)	9.5%	assumption (mid-cap industrial hurdle)
Appraisal horizon	5 years	assumption
Annual O&M	CAD $1,300,000/yr (14% of capex)	assumption: estimate (overtime -40%, penalties -50%; dwell 6.0 to 4.7 days)
Benefit during implementation	30% of full run-rate	assumption (pilot ramp)